

DOPT: D-learning with Off-Policy Target toward Sample Efficiency and Fast Convergence Control

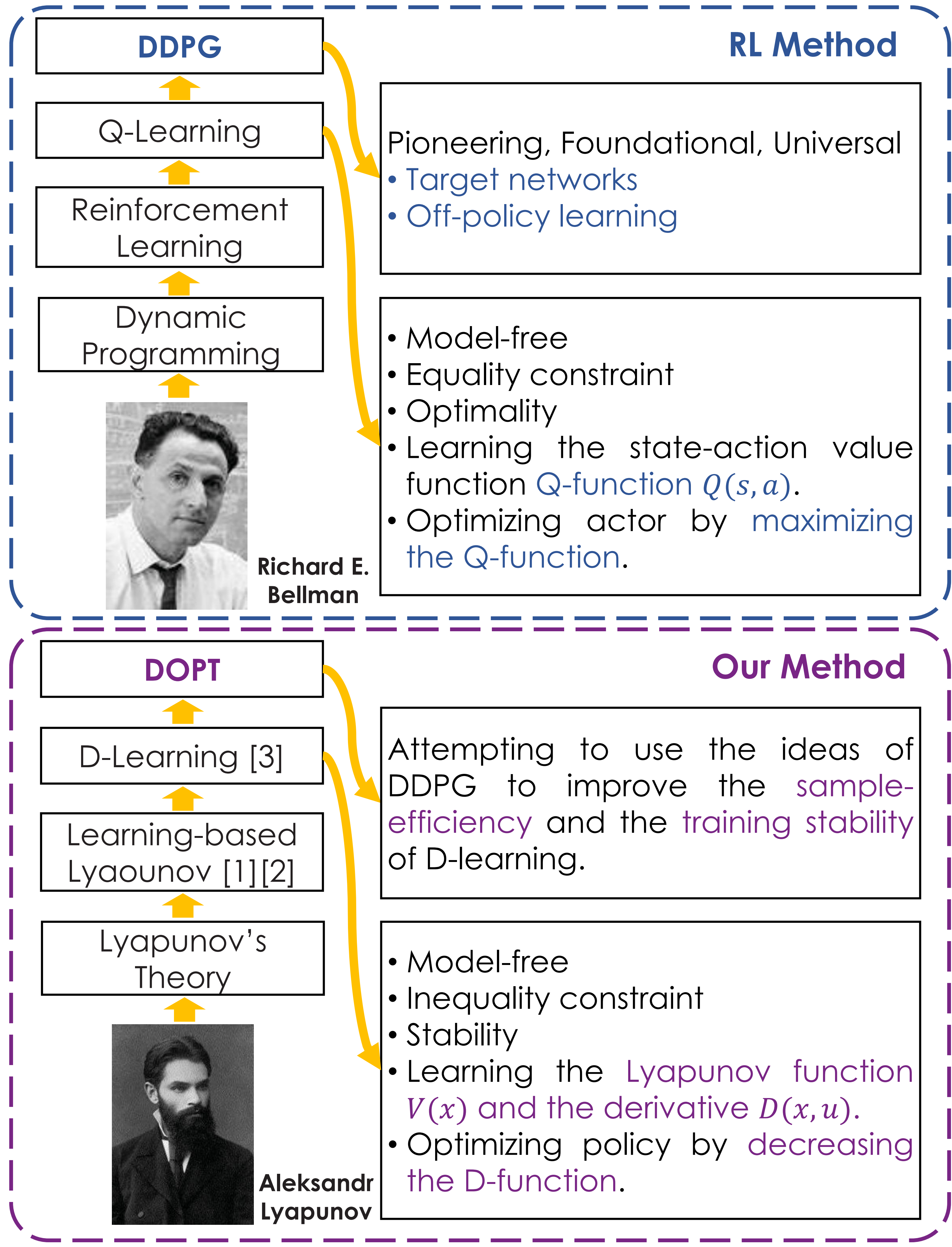
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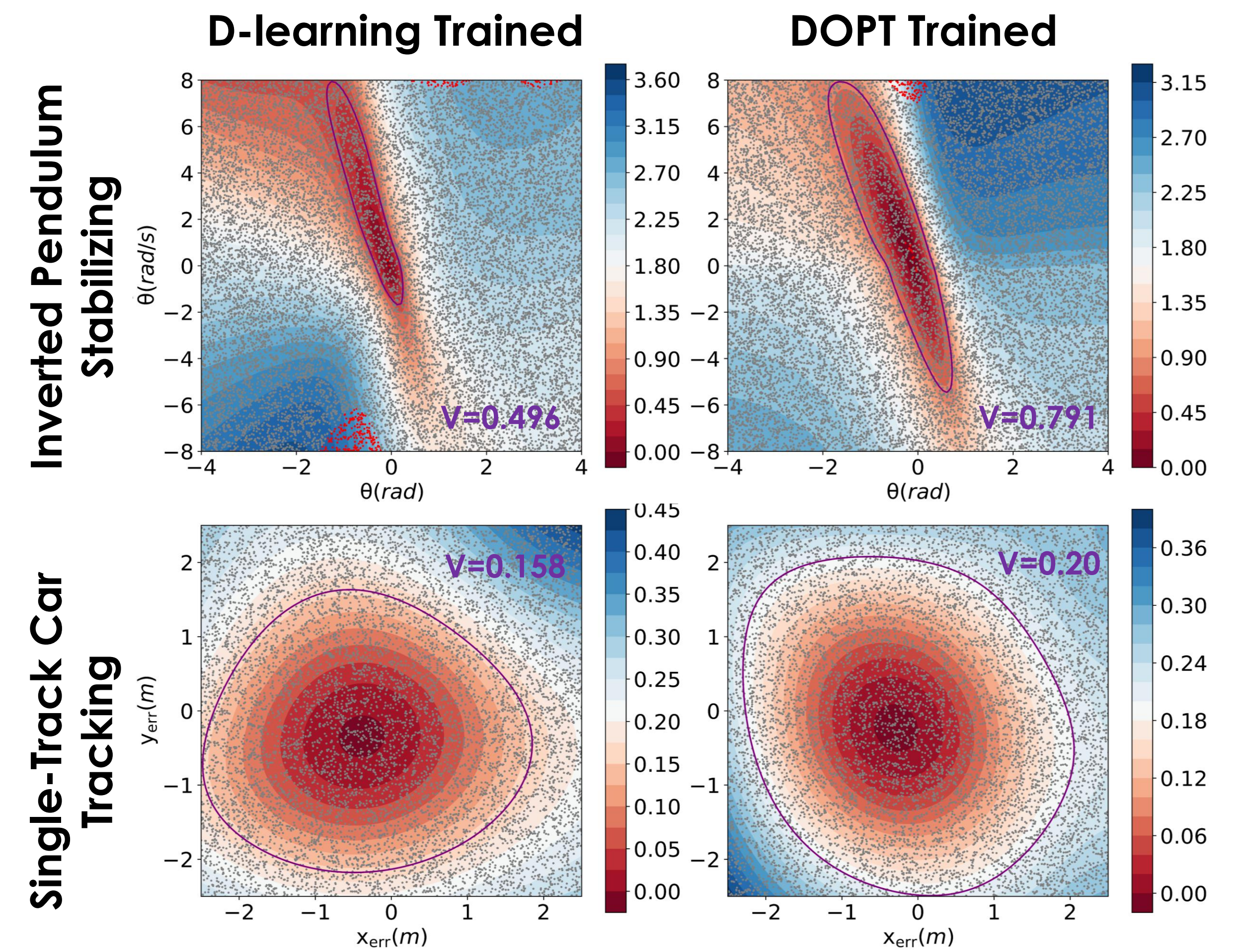


Motivation

Two Parallel Evolutions



Analysis of Lyapunov Candidate and RoA

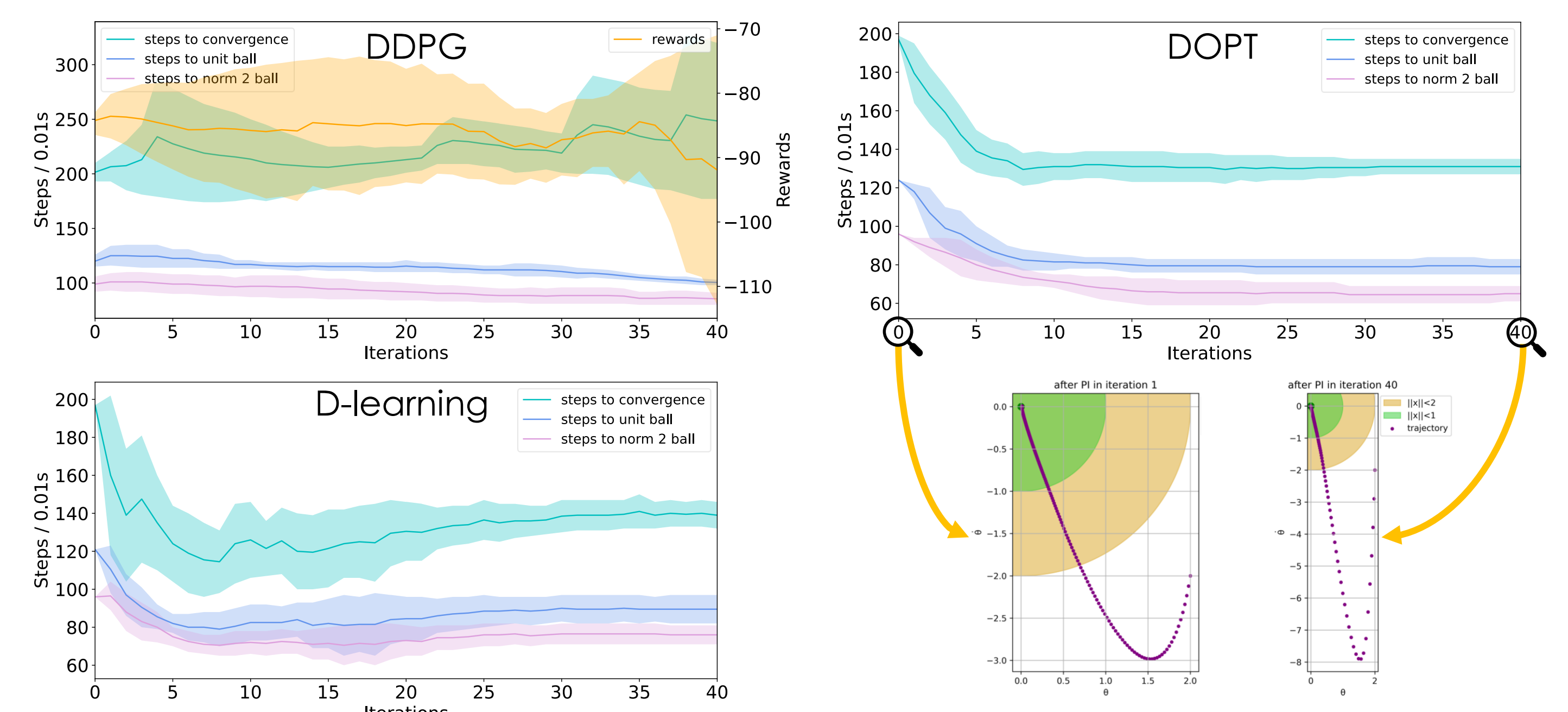


Using sample-based method to verify Lyapunov conditions:

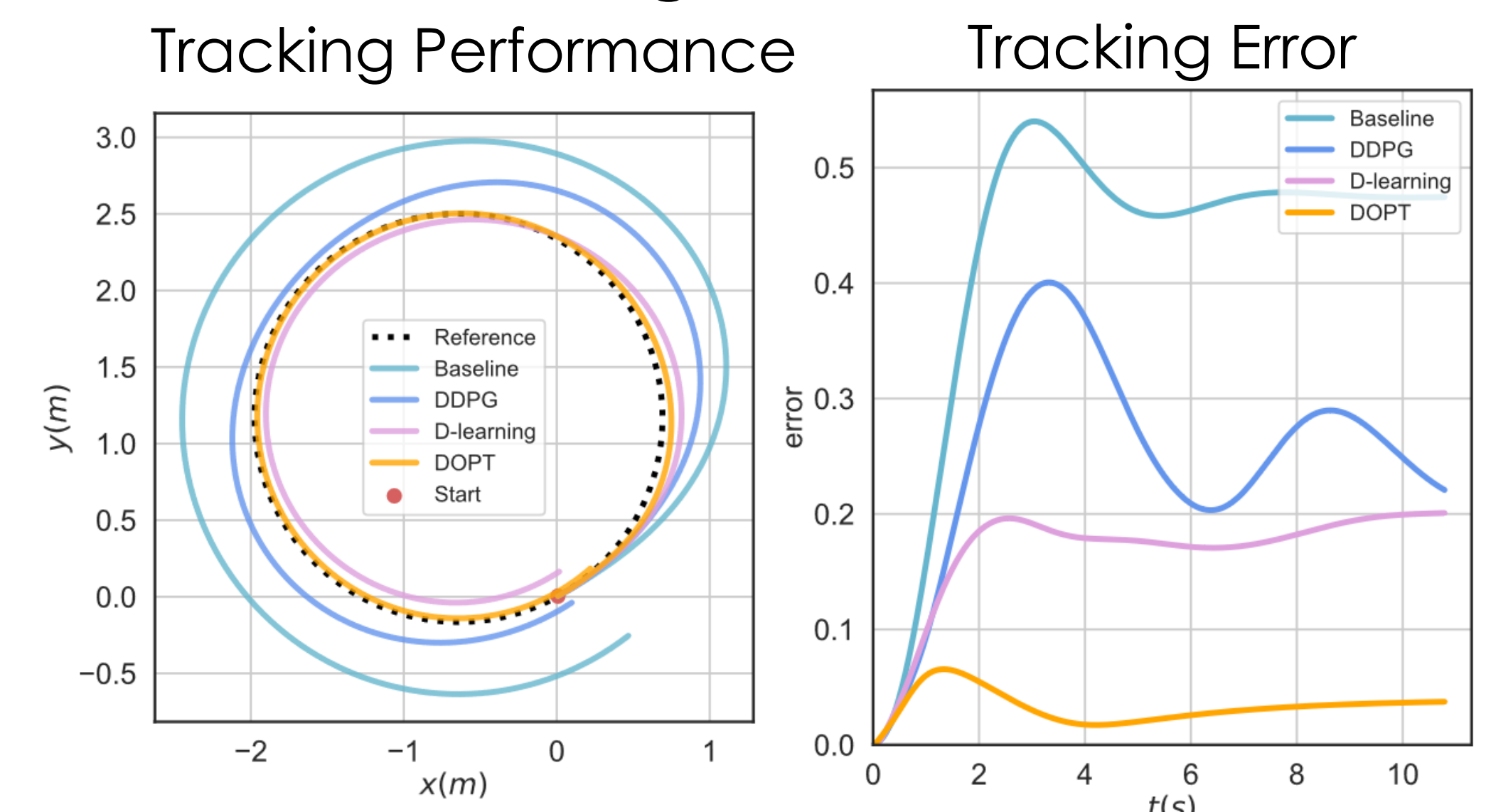
- Lyapunov candidates obtained by D-learning and DOPT satisfy almost Lyapunov conditions [4].
- DOPT can capture more expressive Lyapunov candidates, and provide expanded estimated RoAs.

Simulation

Inverted Pendulum Stabilizing



Single-Track Car Tracking

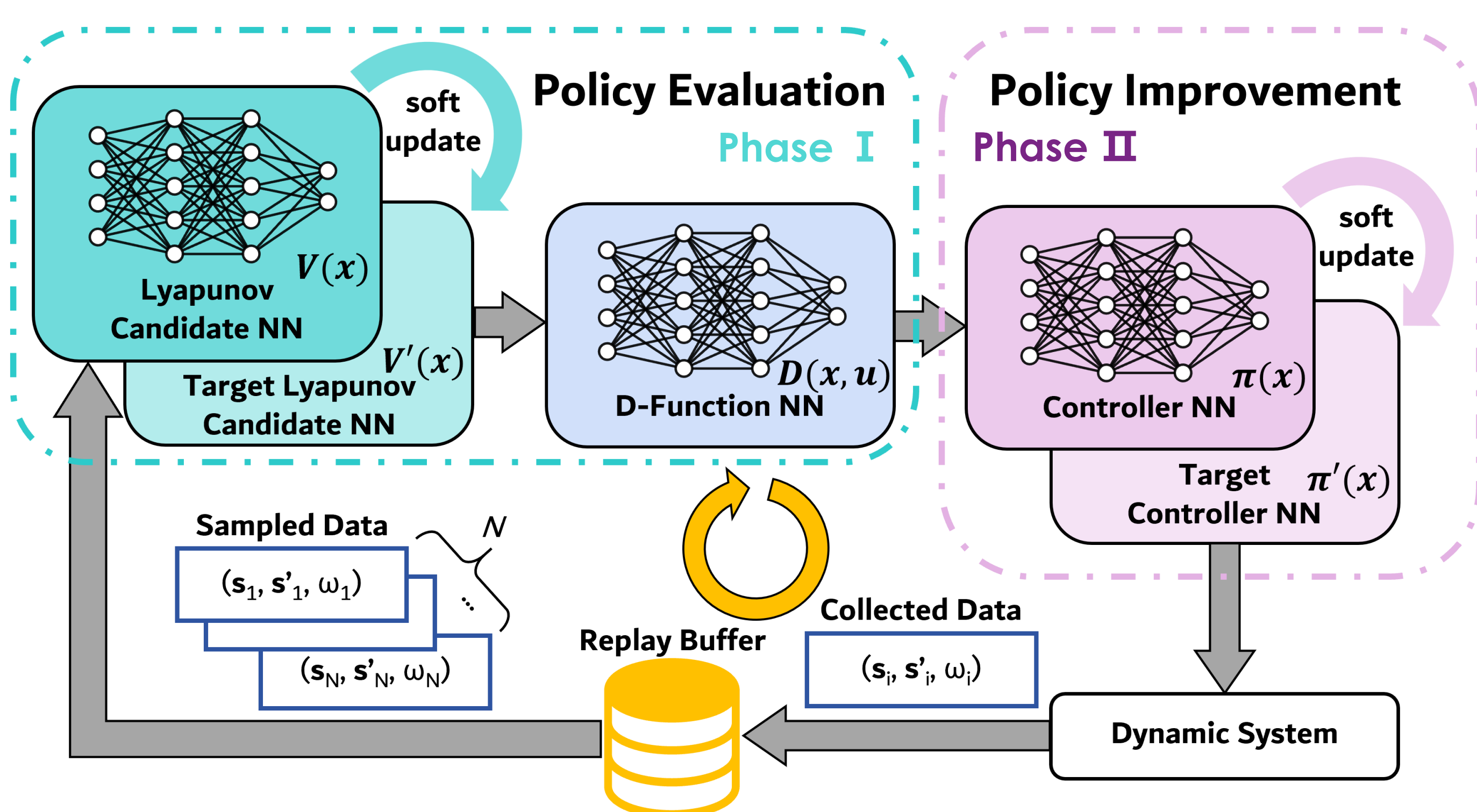


Efficiency Comparison

Approaches	Environment	Data volume (sample/training)	Training time	Iterations	Final PII	Indicator
D-learning	inverted pendulum	2000/2000	31min46s	40	-610.296	144 (convergence steps)
DOPT	inverted pendulum	1000/2000	28min21s	40	-624.529	136
DOPT	inverted pendulum	2000/4000	34min29s	40	-669.143	122
D-learning	single-track car	2000/2000	63min47.9s	20	-98.4	0.152 (steady-state error)
DOPT	single-track car	1000/2000	52min1.6s	20	-110.5	0.124
DOPT	single-track car	2000/4000	69min48.9s	20	-113.7	0.085

Steadier training process, faster convergence, lower tracking error and higher sample efficiency.

Proposed Approach



Initialization

- Initialize a controller $\pi_\phi(x)$ with any admissible (i.e. stabilizing) parameter.

Data Collection

- Collect weighted data (x, x', ω) from closed-loop system $f(x, \pi_\phi(x))$.
- Store weighted data in the Replay Buffer.

Policy Evaluation Phase I

- Sample from the Replay Buffer.
- Train $V_\theta(x)$ using weighted Lyapunov constrains.
- Soft update the target Lyapunov candidate $V'_{\theta_t}(x)$.
- Train D-function $D_\delta(x, u)$ using $V'_{\theta_t}(x') - V'_{\theta_t}(x)$.

Policy Improvement Phase II

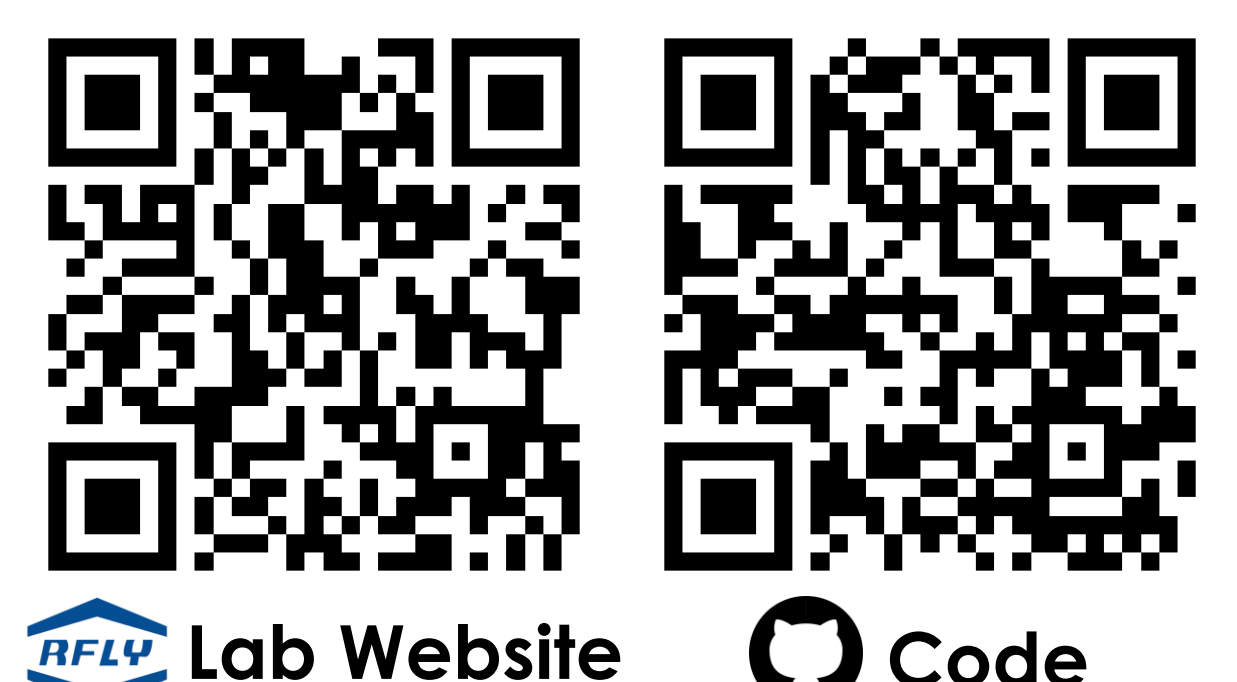
- Improve $\pi_\phi(x)$ by decreasing $D_\delta(x, \pi_\phi(x))$.
- Soft update the target controller $\pi'_{\phi_t}(x)$.

Reference

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Lab Website

Code